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2014.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - V8 S/C 5.0L PETROL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the fuel charging and controls system and operation, refer to the relevant Description and Operation section of the workshop manual.

REFER to: Fuel Charging and Controls (303-04E, Description and Operation).

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.



NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Fuel level ▪ Fuel leaks ▪ Damaged fuel lines ▪ Damaged push connect fittings ▪ Fuel contamination/grade/quality ▪ Throttle body ▪ Damaged fuel tank filler pipe cap ▪ Damaged fuel tank filler pipe	▪ Fuses ▪ Loose or corroded electrical connectors ▪ Harnesses ▪ Sensor(s) ▪ Engine Control Module (ECM)

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Engine cranks, but does not fire	▪ Engine breather system disconnected /restricted ▪ Ignition system ▪ Fuel system ▪ Electronic engine control	Ensure the engine breather system is free from restriction and is correctly installed. Check for ignition system, fuel system and electronic engine control DTCs and refer to the relevant DTC Index
Engine cranks and fires, but will not start	▪ Evaporative emissions purge valve ▪ Fuel pump ▪ Spark plugs ▪ HT short to ground (tracking) check rubber boots for cracks /damage ▪ Ignition system	Check for evaporative emissions, fuel system and ignition system related DTCs and refer to the relevant DTC Index
Difficult cold start	▪ Engine coolant level /anti-freeze content ▪ Battery ▪ Electronic engine controls ▪ Fuel pump ▪ Purge valve	Check the engine coolant level and condition. Ensure the battery is in a fully charged and serviceable condition. Check for electronic engine controls, engine emissions, fuel system and evaporative emissions system related DTCs and refer to the relevant DTC Index

Difficult hot start	▪ Injector leak ▪ Electronic engine control ▪ Purge valve ▪ Fuel pump ▪ Ignition system	Check for injector leak, install new injector as required. Check for electronic engine controls, evaporative emissions, fuel system, ignition system and engine emission system related DTCs and refer to the relevant DTC Index
Difficult to start after hot soak (vehicle standing, engine off, after engine has reached operating temperature)	▪ Injector leak ▪ Electronic engine control ▪ Purge valve ▪ Fuel pump ▪ Ignition system	Check for injector leak, install new injector as required. Check for electronic engine controls, evaporative emissions, fuel system, ignition system and engine emission system related DTCs and refer to the relevant DTC Index
Engine stalls soon after start	▪ Breather system disconnected /restricted ▪ ECM relay ▪ Electronic engine control ▪ Ignition system ▪ Air intake system restricted ▪ Air leakage ▪ Fuel lines	Ensure the engine breather system is free from restriction and is correctly installed. Check for electronic engine control, ignition system and fuel system related DTCs and refer to the relevant DTC Index. Check for blockage in air filter element and air intake system. Check for air leakage in air intake system
Engine hesitates /poor acceleration	▪ Fuel pressure, fuel pump, fuel lines ▪ Injector leak ▪ Air leakage ▪ Electronic engine control ▪ Throttle motor ▪ Restricted accelerator pedal travel (carpet, etc)	Check for fuel system related DTCs and refer to the relevant DTC Index. Check for injector leak, install new injector as required. Check for air leakage in air intake system. Ensure accelerator pedal is free from restriction. Check for electronic engine controls, ignition, engine emission system and transmission related DTCs and refer to the relevant DTC Index

	▪ Ignition system ▪ Transmission malfunction	
Engine backfires	▪ Fuel pump /lines ▪ Air leakage ▪ Electronic engine controls ▪ Ignition system ▪ Sticking variable camshaft timing (VCT) hub	Check for fuel system failures. Check for air leakage in intake air system. Check for electronic engine controls, ignition system and VCT system related DTCs and refer to the relevant DTC Index
Engine surges	▪ Fuel pump /lines ▪ Electronic engine controls ▪ Throttle motor ▪ Ignition system	Check for fuel system failures. Check for electronic engine controls, throttle system and ignition system related DTCs and refer to the relevant DTC Index
Engine detonates /knocks	▪ Fuel pump /lines ▪ Air leakage ▪ Electronic engine controls ▪ Sticking VCT hub	Check for fuel system failures. Check for air leakage in intake air system. Check for electronic engine controls and VCT system related DTCs and refer to the relevant DTC Index
No throttle response	▪ Electronic engine controls ▪ Throttle motor	Check for electronic engine controls and throttle system related DTCs and refer to the relevant DTC Index
Poor throttle response	▪ Breather system disconnected /restricted ▪ Electronic engine control	Ensure the engine breather system is free from restriction and is correctly installed. Check for electronic engine controls, transmission and traction control related DTCs and refer to the related DTC Index. Check for air leakage in intake air system

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| | <ul style="list-style-type: none">▪ Transmission malfunction▪ Traction control event▪ Air leakage | |
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DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code (DTC) Index - V8 S/C 5.0L Petrol, DTC: Engine Control Module (ECM) (100-00, Description and Operation).